

Product Requirement Document

App Name: CareerGuid AI

Alternative name: **Skillsight**

1. Product Overview

CareerGuid AI is an AI-powered career guidance platform that helps students, freshers, and job seekers analyze their resume, identify skill gaps, generate personalized learning roadmaps, and build ATS-friendly resumes.

The platform uses **AI/ML, RAG pipeline, GenAI agent, resume parsing, system design, Supabase PostgreSQL, Supabase Storage, Redis, rate limiting, OTP verification, Docker deployment, queue-based notifications, and scalable backend architecture.**

The main purpose of this app is to help users answer:

“Mere current skills ke basis par mujhe target job role ke liye kya-kya improve karna chahiye?”

2. Vision Statement

To build a smart AI career assistant that helps users become job-ready by analyzing their resume, identifying missing skills, suggesting learning paths, sending progress reminders, and generating ATS-friendly resumes for job applications.

3. Problem Statement

Many students and freshers face these problems:

1. Unhe pata nahi hota ki resume job role ke according strong hai ya nahi.
2. Resume me skills hain, lekin target job role ke required skills se match nahi hoti.
3. ATS-friendly resume banana difficult hota hai.
4. Learning roadmap clear nahi hota.
5. Interview preparation ke liye correct topics identify nahi kar pate.
6. Manual profile creation boring aur time-consuming hoti hai.
7. Users ko pata nahi hota ki unka resume me email, phone, skills, projects, experience properly mention hai ya nahi.
8. Users roadmap generate kar lete hain, lekin timely complete nahi kar pate.

CareerGuid AI in problems ko solve karega using AI-based resume analysis, resume-first smart onboarding, skill gap detection, RAG-based roadmap, progress tracking, weekly email reminders, and ATS resume builder.

4. Target Users

Primary Users

User Type	Description
Students	College students looking for internships or fresher jobs
Freshers	Recent graduates applying for entry-level roles
Job Seekers	Candidates switching roles or preparing for interviews
Bootcamp Learners	People learning tech skills and building job-ready profiles

Secondary Users

User Type	Description
Career Coaches	Can use platform to guide students
Training Institutes	Can track student skill gaps
Colleges	Can help students improve employability
Recruiters, future version	Can view verified candidate profiles

5. User Roles

5.1 Guest User

A guest user can:

- Visit landing page
- View app overview
- View sample roadmap
- View pricing, if added later
- Register or login

5.2 Registered User

A registered user can:

- Create account
- Verify email using OTP
- Upload resume during onboarding
- Auto-fill profile from resume
- Edit incorrect auto-filled details
- Manually fill missing details
- Select target job role
- Analyze skill gap
- Generate RAG-based roadmap
- Track roadmap progress
- Receive weekly reminder emails if roadmap tasks are pending
- Build ATS-friendly resume
- Download resume as PDF/DOCX

5.3 Admin

Admin can:

- Manage users
- Manage job roles
- Manage required skills
- Manage roadmap knowledge base

- Manage AI prompt templates
- View analytics
- Monitor failed resume parsing jobs
- View reminder logs
- View system logs

5.4 AI System / Agent

AI agent can:

- Parse resume
 - Extract skills
 - Auto-fill user profile
 - Match skills with job role
 - Generate learning roadmap
 - Suggest ATS keywords
 - Improve project descriptions
 - Generate resume summary
 - Recommend next learning steps
 - Help track roadmap progress
-

6. Core Product Modules

6.1 Authentication Module

Features:

- Register
- Login
- Email verification using OTP
- Forgot password
- Reset password
- JWT authentication
- Password hashing using bcrypt
- Rate limiting using Redis

Authentication Flow

```
User registers
  ↓
System validates input
  ↓
OTP generated
  ↓
OTP stored in Redis with expiry
  ↓
OTP sent through email
  ↓
User verifies OTP
  ↓
Account activated
```

6.2 Smart Onboarding Module

When a user enters the app for the first time, the system will first ask the user to **upload their resume**.

The system will use AI-based resume parsing to automatically extract user details from the uploaded resume and fill the onboarding form.

After auto-fill, the user can review all details. If some details are missing, the user can manually add them. If any auto-filled detail is incorrect, the user can edit and update it before final submission.

Smart Onboarding Flow

```
graph TD
    A[User enters app for the first time] --> B[Resume upload screen appears]
    B --> C[User uploads PDF/DOCX resume]
    C --> D[AI Resume Parser extracts details]
    D --> E[System auto-fills onboarding form]
    E --> F[User reviews auto-filled details]
    F --> G[User edits incorrect details or fills missing details]
    G --> H[User submits final profile]
    H --> I[Profile saved in Supabase PostgreSQL]
    I --> J[User redirected to dashboard]
```

Auto-Filled Details from Resume

The system will try to automatically extract and fill:

- Full name
- Email
- Phone number
- Current city
- Education
- Work experience
- Skills
- Projects
- LinkedIn URL
- GitHub URL
- Portfolio URL

Manually Filled or Edited Details

Some details may not be available in the resume, so the user can manually fill them:

- Target job role
- Preferred location

Work preference: Remote, Hybrid, Onsite
Expected salary
Notice period
Career goal

Also, the user can edit any auto-filled field if the extracted data is incorrect.

Field	Source	User Action
Full Name	Auto-filled from resume	User can edit
Email	Auto-filled from resume	User can edit
Phone Number	Auto-filled from resume	User can edit
Skills	Auto-filled from resume	User can add/remove skills
Projects	Auto-filled from resume	User can edit project details
LinkedIn URL	Auto-filled if found	User can edit/add
GitHub URL	Auto-filled if found	User can edit/add
Portfolio URL	Auto-filled if found	User can edit/add
Target Job Role	Manual	User selects role
Preferred Location	Manual	User enters location
Work Preference	Manual	User selects remote/hybrid/onsite
Expected Salary	Manual	User enters expected salary
Notice Period	Manual	User enters notice period

Updated Feature Explanation

In this module, the onboarding process becomes resume-first. Instead of asking the user to manually fill all details first, the system first asks the user to upload a resume.

The AI resume parser extracts important information such as name, email, phone number, education, skills, experience, projects, LinkedIn, GitHub, and portfolio links.

After extraction, the form is automatically filled. The user only needs to review the details, correct any wrong information, and complete the missing fields like target job role, preferred location, work preference, expected salary, or notice period.

This reduces manual effort, saves time, and improves the onboarding experience.

User Story

As a first-time user, I want to upload my resume first, so that the system can automatically fill my profile details and I only need to edit or complete the missing information.

Acceptance Criteria

User should see resume upload option first during onboarding.
User should be able to upload PDF/DOCX resume.
System should extract details from the resume.
System should auto-fill the onboarding form.
User should be able to edit auto-filled details.
User should be able to fill missing details manually.
User should be able to submit the final profile.
Final verified profile should be saved in Supabase PostgreSQL.
After successful onboarding, user should be redirected to the dashboard.

Final Short Version for PRD

Smart Onboarding Module: During first-time onboarding, the user will first upload their resume. The system will parse the resume using AI and automatically fill details such as name, email, phone number, education, skills, work experience, projects, LinkedIn, GitHub, and portfolio. After auto-fill, the user can review the information, edit incorrect fields, and manually complete missing details such as target job role, preferred location, work preference, expected salary, and notice period. Once reviewed, the user submits the final profile, which is stored in Supabase PostgreSQL.

6.3 Resume Parsing Module

The system should support:

- PDF resume upload
 - DOCX resume upload
 - File size validation
 - File type validation
 - Resume text extraction
 - Email extraction
 - Phone extraction
 - Skill extraction
 - Education extraction
 - Experience extraction
 - Project extraction
 - Certification extraction
 - LinkedIn/GitHub/Portfolio extraction
-

6.4 Skill Gap Analysis Module

The user selects a target role.

Example roles:

- Backend Developer
- Frontend Developer
- Full Stack Developer
- Data Analyst
- AI/ML Engineer
- Cloud Engineer
- DevOps Engineer
- Java Developer
- MERN Stack Developer

The system compares:

User current skills

vs

Target role required skills

Output:

- Current skills
- Missing skills
- Recommended skills
- Match score
- Priority score

- Suggested learning order

Example:

Target Role: Backend Developer

Current Skills:

Java, SQL, HTML, CSS

Missing Skills:

Node.js, Express.js, MongoDB, REST API, JWT, Redis, System Design

Match Score:

58%

6.5 RAG-Based Career Guidance and Roadmap Generation Module

RAG means **Retrieval Augmented Generation**.

The app will use a knowledge base containing:

- Job role requirements
- Skill descriptions
- Learning resources
- Interview questions
- Roadmaps
- Project ideas
- ATS keywords
- Resume writing tips

RAG Flow:

User profile + target role + skill gap

↓

Convert query into embedding

↓

Search Supabase pgvector index

↓

Retrieve relevant career data

↓

Send data to GenAI model

↓

Generate personalized roadmap

RAG-Based Roadmap Progress Reminder

When the GenAI/RAG system generates a personalized roadmap, the roadmap will be divided into week-wise tasks.

Example:

Week 1: JavaScript basics

Week 2: Node.js and Express.js

Week 3: MongoDB and REST API

Week 4: JWT Authentication

Week 5: Backend Project
Week 6: System Design Basics

The user can mark tasks as completed from the dashboard. If the user is not completing tasks on time or a weekly task is pending, the system will send a **weekly email reminder**.

Roadmap Reminder Flow

```
RAG generates personalized roadmap
↓
Roadmap saved in Supabase PostgreSQL
↓
User gets weekly tasks
↓
System checks progress every week
↓
If task is pending
↓
Email reminder sent to user
↓
User comes back and completes task
```

Reminder Logic

```
If currentWeekTask.status != completed:
    Send weekly reminder email

If user inactive for 7 days:
    Send progress reminder email

If roadmap progress < expected progress:
    Send motivational email + pending tasks
```

Example email:

Subject: Your Backend Developer roadmap is pending

Hi Nitin,

You have not completed your Week 2 tasks yet.

Pending topics:

- Node.js basics
- Express.js routing
- REST API practice

Complete these tasks to stay on track for your Backend Developer goal.

New Features Added in Roadmap Module

Personalized roadmap generation
Week-wise learning plan
Task completion tracking
Roadmap progress percentage

Weekly email reminders
Inactive user reminder
Pending task reminder
Motivational progress email

Components Required

1. Roadmap Service

Responsible for:

Generating roadmap
Saving roadmap
Creating weekly milestones
Creating tasks
Updating task status

2. Progress Tracking Service

Responsible for:

Track completed tasks
Track pending tasks
Calculate progress percentage
Check if user is behind schedule

3. Scheduler / Cron Job

Responsible for:

Run weekly check
Find users with pending roadmap tasks
Create reminder email jobs

4. Notification Service

Responsible for:

Send weekly reminder email
Send pending task email
Send motivational email

5. Message Queue

Use queue for async email sending.

Cron Job → Queue → Email Worker → Mail Sent

This avoids blocking the main backend server.

6.6 GenAI Career Agent

The GenAI agent will behave like a personal career mentor.

It will:

- Understand user profile

- Analyze current skills
- Understand target role
- Prioritize missing skills
- Generate weekly roadmap
- Recommend projects
- Recommend interview topics
- Suggest resume improvements
- Suggest next step after every milestone
- Help user stay consistent through roadmap reminders

Example output:

Week 1: JavaScript basics and Git
 Week 2: Node.js and Express.js
 Week 3: MongoDB and REST API
 Week 4: Authentication using JWT
 Week 5: Build backend project
 Week 6: Learn system design basics

6.7 ATS-Friendly Resume Builder

The user can generate a professional resume using AI.

Features:

- AI resume summary generation
- Skills section optimization
- Project description improvement
- ATS keyword suggestions
- Role-based resume tailoring
- Grammar improvement
- Resume preview
- PDF download
- DOCX download

Flow:

```

User profile + target role
    ↓
AI suggests keywords
    ↓
AI generates resume content
    ↓
User previews resume
    ↓
User edits if needed
    ↓
Download as PDF/DOCX
  
```

6.8 Dashboard Module

The dashboard will show:

- Profile completion percentage
- Resume match score
- Target role

- Missing skills
 - Weekly roadmap
 - Pending roadmap tasks
 - Completed roadmap tasks
 - Roadmap progress percentage
 - Recommended projects
 - ATS resume score
 - Learning progress
 - Recent AI suggestions
-

6.9 Notification Module

Notifications include:

- Welcome email
- OTP email
- Password reset email
- Roadmap generated notification
- Resume analysis completed notification
- Weekly roadmap reminder email
- Pending task reminder email
- Inactive user reminder email
- Motivational progress email

For high-level system design, email sending should be async.

User action / Cron job

↓

Backend creates email job

↓

Job pushed to queue

↓

Worker sends email

7. System Design Requirements

7.1 High-Level Architecture

User

↓

React Frontend

↓

Load Balancer / Nginx

↓

API Gateway

↓

Backend Services

- Auth Service
- Onboarding Service
- Resume Upload Service
- Resume Parsing Service
- Skill Gap Service
- RAG Service
- Roadmap Service

- Progress Tracking Service
- ATS Resume Builder Service
- Notification Service

Supporting Systems:

- Supabase PostgreSQL
 - Supabase Storage
 - Supabase pgvector for RAG embeddings
 - Redis
 - Message Queue
 - Scheduler / Cron Job
 - AI/ML Service
-

7.2 Supabase Usage

Supabase will replace MongoDB as the primary database layer. The project will use Supabase for:

- Supabase PostgreSQL for users, profiles, resumes, skill analysis, roadmaps, progress, and reminder logs
 - Supabase Storage for uploaded resumes and generated PDF/DOCX resumes
 - Supabase pgvector for RAG embeddings and semantic retrieval
 - Row Level Security policies to protect user-specific profile and roadmap data
-

7.3 Redis Usage

Redis will be used for:

- OTP storage
- Forgot password token
- Rate limiting
- Temporary session cache
- AI response caching
- Roadmap cache
- Queue support with BullMQ

Example:

```
otp:user@gmail.com = 459201  
expiry = 5 minutes
```

7.4 Rate Limiting

Rate limiting should be applied in two layers: **system-level throughput** and **anti-abuse protection**. The login API should not be limited globally to a very low number. The platform must support at least **100 users per minute logging in successfully** under normal traffic.

System-Level Throughput Requirement

API / Flow	Minimum Capacity	Purpose
Login API	100 successful user logins/minute system-wide	Allows multiple users to login at the same time without blocking genuine users
Register API	50 registrations/minute system-wide	Supports user growth during campaigns or college drives
Resume Upload API	100 uploads/hour system-wide	Controls heavy file uploads while supporting normal usage
AI Roadmap Generate	20 generations/day per user	Controls AI cost and prevents misuse
ATS Resume Generate	10 generations/day per user	Controls AI/PDF generation load

Anti-Abuse Rate Limits

API / Action	Limit	Applied On
Failed Login Attempts	5 failed attempts/minute	Per IP + per email
Successful Login Requests	20 requests/minute	Per IP
Send OTP	3 requests/10 minutes	Per email + per IP
Forgot Password	3 requests/15 minutes	Per email + per IP
Resume Upload	10 uploads/hour	Per user
AI Roadmap Generate	20 requests/day	Per user
ATS Resume Generate	10 requests/day	Per user

This design allows **minimum 100 users per minute to login**, while still protecting the system from brute-force attacks, OTP spam, and AI misuse using Redis-based counters.

7.5 Load Balancer

Load balancer will distribute traffic between backend instances.

Request



Load Balancer

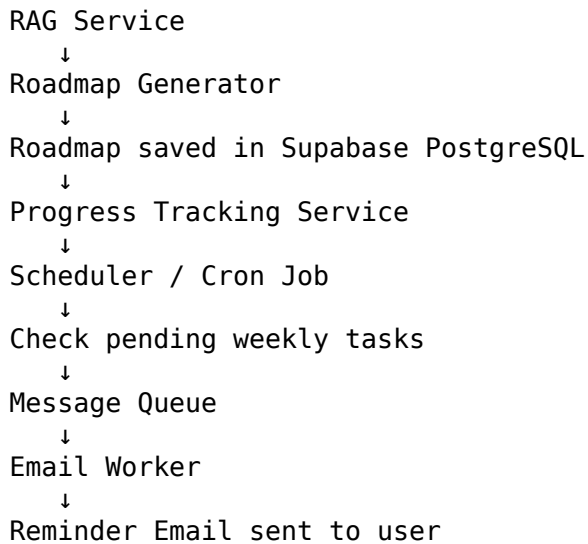


Backend 1 / Backend 2 / Backend 3

Benefits:

- Better scalability
- High availability
- No single backend overload
- Easy horizontal scaling

7.6 Roadmap Reminder System Design



7.7 Docker Deployment

The app should be containerized using Docker.

Containers:

frontend-container
backend-container
ai-service-container
redis-container
nginx-container
worker-container

Supabase will be used as the managed backend database layer, so a MongoDB container is not required. For local development, developers can either connect to a Supabase cloud project or use the Supabase local development stack. Docker Compose will be used for the application services and VPS deployment.

8. Feature Categorization by Version

Version 0.1 — Prototype / Proof of Concept

Goal: Basic working demo.

Features

- Landing page
- Register/login UI
- Basic dashboard UI
- Resume upload UI
- Static skill gap result
- Static roadmap result
- Basic profile form

Purpose

This version is only for validating the idea and showing the UI flow.

Version 1.0 — MVP Release

Goal: Launch first usable product.

Category	Features
Authentication	Register, login, JWT authentication
Email Verification	OTP generation, OTP verification using Redis
Smart Onboarding	Resume-first onboarding, resume upload, auto-filled form, manual completion
Resume Parsing	Extract name, email, phone, skills, education, experience, projects
Profile Auto-Fill	Auto-fill details from resume
User Editing	User can edit incorrect details and fill missing fields
Skill Gap	Compare user skills with selected target role
Dashboard	Show profile, missing skills, match score
Basic Roadmap	Generate simple learning roadmap
Security	Password hashing, input validation
Deployment	Docker Compose local deployment with Supabase integration

Version 1.1 — ATS Resume Builder Release

Goal: Help users create job-ready resumes.

Category	Features
Resume Builder	AI-generated resume summary
ATS Optimization	Role-based keywords
Resume Editing	User can edit generated content
Download	PDF download
Resume Preview	Preview before download
Project Improvement	AI improves project descriptions

Version 1.2 — RAG and GenAI Roadmap Release

Goal: Make guidance more personalized and intelligent.

Category	Features
RAG Pipeline	Supabase pgvector integration
Knowledge Base	Job roles, skills, roadmaps, interview topics
AI Agent	Personalized roadmap generation
Roadmap Structure	Week-wise learning plan
Task Structure	Task-based roadmap structure
Dashboard	Roadmap saved in dashboard
Progress Basics	Basic task completion status
Chat Assistant	Ask career-related questions
Project Recommendation	AI suggests projects based on role
Interview Prep	AI suggests interview topics

Version 1.3 — Scalability and Security Release

Goal: Make system production-ready.

Category	Features
Load Balancing	Nginx / cloud load balancer
Rate Limiting	Redis-based API rate limiting with minimum 100 successful logins/minute system-wide
Queue System	Async email and AI jobs
Monitoring	Logs, error tracking
File Security	File type and size validation
Forgot Password	OTP/token-based reset
Welcome Email	Async notification
AI Caching	Cache repeated roadmap responses

Version 1.4 — Progress Tracking and Reminder Release

Goal: Help users track learning journey and stay consistent.

Category	Features
Progress Tracker	Mark roadmap tasks complete
Weekly Plan	Week-wise learning plan
Roadmap Progress	Progress percentage
Reminder System	Weekly reminder emails
Pending Task Notification	Email if weekly task is pending
Inactive User Reminder	Email if user inactive for 7 days
Motivational Email	Motivation + pending tasks
Reminder Logs	Store email reminder history
Dashboard Analytics	Profile score, ATS score, roadmap progress

Version 2.0 — Advanced Career Platform

Goal: Convert product into a full career platform.

Category	Features
Recruiter Portal	Recruiters can view candidate profiles
College Dashboard	College can track student readiness
AI Mock Interview	Practice interviews with AI
Job Matching	Match users with job postings
Multi-Resume Support	Different resumes for different roles
Advanced ATS Score	Detailed resume scoring
Mobile App	Android/iOS app
Payment Plans	Premium features

9. User Stories

Authentication

User Story 1

As a new user, I want to register using my email, so that I can create an account on the platform.

Acceptance Criteria

- User can enter name, email, and password
 - Password is stored securely
 - OTP is sent to email
 - User must verify email before using dashboard
-

Smart Onboarding

User Story 2

As a first-time user, I want to upload my resume first, so that the system can automatically fill my profile details and I only need to edit or complete the missing information.

Acceptance Criteria

- User should see resume upload option first during onboarding
 - User should be able to upload PDF/DOCX resume
 - System should extract details from the resume
 - System should auto-fill the onboarding form
 - User should be able to edit auto-filled details
 - User should be able to fill missing details manually
 - User should be able to submit the final profile
 - Final verified profile should be saved in Supabase PostgreSQL
 - After successful onboarding, user should be redirected to dashboard
-

Skill Gap Analysis

User Story 3

As a job seeker, I want to select a target job role, so that I can know which skills I am missing.

Acceptance Criteria

- User can select target role
 - System fetches required skills
 - System compares with user skills
 - User sees current skills, missing skills, and match score
-

Roadmap Generation

User Story 4

As a learner, I want a personalized roadmap, so that I know what to study step by step.

Acceptance Criteria

- Roadmap is generated based on missing skills
 - Roadmap is divided week-wise
 - Roadmap includes topics and project suggestions
 - Roadmap is saved in dashboard
-

Roadmap Reminder

User Story 5

As a user, I want to receive weekly reminders when I do not complete my roadmap tasks, so that I can stay consistent with my learning plan.

Acceptance Criteria

- System should generate a week-wise roadmap
 - User should be able to mark tasks as completed
 - System should track roadmap progress
 - System should check pending tasks weekly
 - If weekly tasks are incomplete, system should send reminder email
 - Reminder email should include pending topics
 - System should not send duplicate reminders for the same week repeatedly
 - User should be able to continue roadmap from dashboard
-

ATS Resume Builder

User Story 6

As a job seeker, I want to generate an ATS-friendly resume, so that I can apply to jobs with a better resume.

Acceptance Criteria

- User can generate resume using profile data
 - AI suggests keywords
 - AI improves summary and project descriptions
 - User can preview resume
 - User can download resume as PDF
-

Forgot Password

User Story 7

As a user, I want to reset my password using OTP, so that I can recover my account securely.

Acceptance Criteria

- User enters email
- OTP/token sent to email
- OTP stored in Redis with expiry
- User verifies OTP
- User sets new password

- Old password is replaced with hashed new password
-

10. Functional Requirements

Authentication

- User should be able to register
- User should be able to login
- System should send OTP
- System should verify OTP
- User should be able to reset password
- JWT should protect private APIs

Onboarding

- User should upload resume first
- System should extract resume data
- System should auto-fill available fields
- User should edit incorrect auto-filled fields
- User should manually complete missing fields
- System should save final profile

Resume Analysis

- System should parse resume
- System should extract skills
- System should normalize skills
- System should compare with job role
- System should generate match score

Roadmap

- System should generate personalized roadmap
- Roadmap should be saved
- User should see roadmap in dashboard
- Roadmap should be divided week-wise
- User should be able to mark tasks as completed
- System should calculate progress percentage

Reminder System

- System should check roadmap progress weekly
- System should identify pending roadmap tasks
- System should send reminder emails
- System should send inactive user emails
- System should store reminder logs
- System should avoid duplicate reminders for the same week

ATS Resume Builder

- System should generate resume content
- System should suggest keywords
- User should preview resume

- User should download resume
-

11. Non-Functional Requirements

Performance

- Dashboard should load within 2 seconds
- Resume parsing should complete within reasonable time
- Large AI tasks should run asynchronously
- Cached roadmap should be returned quickly
- Reminder checks should run in background without affecting user APIs

Security

- Password hashing using bcrypt
- JWT authentication
- Redis OTP expiry
- Rate limiting
- Input validation
- File type validation
- File size validation
- HTTPS in production

Scalability

- Backend should support horizontal scaling
- AI service should be separately scalable
- Redis should support cache and rate limiting
- Queue should handle background jobs
- Supabase Storage should store uploaded resumes and generated resume files
- Scheduler and workers should run separately from main API server

Reliability

- Failed resume parsing jobs should be retryable
 - Email jobs should be queued
 - Reminder email jobs should be logged
 - Logs should be maintained
 - Supabase PostgreSQL backup and recovery should be planned
-

12. Success Metrics

Product Metrics

Metric	Target
Signup completion rate	60%+
Email verification success	80%+
Onboarding completion rate	70%+
Resume upload rate	75%+
Skill gap analysis usage	60%+ users

Metric	Target
Roadmap generation usage	50%+ users
Roadmap task completion rate	40%+
Weekly reminder open rate	30%+
ATS resume download rate	40%+ users
User retention after 7 days	25%+

Technical Metrics

Metric	Target
API success rate	99%
Login throughput capacity	Minimum 100 successful logins/minute
Resume parsing success	90%+
OTP delivery success	95%+
Reminder email delivery success	95%+
Average API response time	Under 500ms for normal APIs
AI task failure rate	Below 5%
System uptime	99%

AI Quality Metrics

Metric	Target
Skill extraction accuracy	85%+
Profile auto-fill accuracy	90%+
Roadmap relevance	80%+ positive feedback
ATS resume usefulness	80%+ positive feedback

13. Agile Development Plan

Workflow

The project will follow Agile/Scrum workflow.

```

Product Backlog
  ↓
Sprint Planning
  ↓
To Do
  ↓
In Progress
  ↓
Code Review
  ↓
Testing
  ↓
Done
  ↓
Sprint Demo
  ↓
Retrospective

```

Sprint Plan

Sprint 1: Authentication

Deliverables:

- Register
- Login
- JWT
- OTP verification
- Redis OTP storage
- Forgot password

Sprint 2: Smart Onboarding

Deliverables:

- Resume-first onboarding flow
- Resume upload
- Resume parsing
- Auto-filled profile form
- Edit incorrect details
- Manual missing details completion

Sprint 3: Skill Gap Analysis

Deliverables:

- Target role selection
- Required skills database
- Skill comparison
- Match score
- Dashboard result

Sprint 4: RAG Roadmap

Deliverables:

- Knowledge base
- Supabase pgvector
- RAG retrieval
- GenAI roadmap
- Week-wise roadmap
- Save roadmap

Sprint 5: ATS Resume Builder

Deliverables:

- AI resume generation
- ATS keyword suggestions
- Resume preview
- PDF download
- DOCX download, optional

Sprint 6: Progress Tracking and Reminder System

Deliverables:

- Task completion tracking
- Roadmap progress percentage
- Weekly reminder email
- Inactive user reminder
- Pending task reminder
- Reminder logs
- Cron job / scheduler
- Queue-based email worker

Sprint 7: Scalability and Deployment

Deliverables:

- Docker setup
 - Docker Compose
 - Nginx reverse proxy
 - Redis rate limiting
 - Queue workers
 - Logging
 - Production deployment
-

14. MVP Definition

MVP Goal

The MVP should prove that the platform can:

1. Register and verify users
2. Upload resume first during onboarding
3. Parse resume
4. Auto-fill available details
5. Let users edit incorrect details
6. Let users manually complete missing details
7. Analyze skill gap
8. Generate basic roadmap
9. Show dashboard

MVP Features

- Register/login
- Email OTP verification
- Resume-first smart onboarding
- Resume upload
- Resume parsing
- Profile auto-fill
- Edit auto-filled details
- Manual missing details
- Target role selection
- Skill gap analysis
- Basic roadmap
- Dashboard

Not Included in MVP

- Recruiter portal

- Payment system
 - Mobile app
 - Advanced AI mock interview
 - College dashboard
 - Advanced ATS scoring
 - Advanced analytics
 - Full weekly reminder automation, planned for Version 1.4
-

15. Supabase Database Design Update

The project will use **Supabase PostgreSQL** as the primary database instead of MongoDB. Supabase will also be used for structured relational data, file metadata, storage integration, and optionally RAG embeddings using **pgvector**.

15.1 Supabase Usage

Supabase PostgreSQL -> users, profiles, resumes, roadmaps, progress, reminder logs
 Supabase Storage -> uploaded resumes and generated PDF/DOCX resumes
 Supabase pgvector -> embeddings for RAG knowledge base and roadmap retrieval
 Redis -> OTP, rate limiting, temporary tokens, cache, queue support

15.2 profiles Table

```
CREATE TABLE profiles (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID NOT NULL,
  full_name TEXT,
  email TEXT UNIQUE NOT NULL,
  phone TEXT,
  current_city TEXT,
  education TEXT,
  work_experience TEXT,
  skills TEXT[],
  projects JSONB,
  linkedin_url TEXT,
  github_url TEXT,
  portfolio_url TEXT,
  target_job_role TEXT,
  preferred_location TEXT,
  work_preference TEXT,
  expected_salary TEXT,
  notice_period TEXT,
  career_goal TEXT,
  onboarding_completed BOOLEAN DEFAULT FALSE,
  created_at TIMESTAMP DEFAULT NOW(),
  updated_at TIMESTAMP DEFAULT NOW()
);
```

15.3 resumes Table

```
CREATE TABLE resumes (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID NOT NULL,
  file_url TEXT NOT NULL,
```



```

file_type TEXT,
extracted_text TEXT,
extracted_skills TEXT[],
extracted_profile JSONB,
parsing_status TEXT DEFAULT 'pending',
created_at TIMESTAMP DEFAULT NOW()
);

```

15.4 roadmaps Table

```

CREATE TABLE roadmaps (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID NOT NULL,
  target_role TEXT NOT NULL,
  weeks JSONB NOT NULL,
  progress_percentage INT DEFAULT 0,
  status TEXT DEFAULT 'active',
  created_at TIMESTAMP DEFAULT NOW(),
  updated_at TIMESTAMP DEFAULT NOW()
);

```

Example weeks JSONB value:

```

[
  {
    "weekNumber": 1,
    "title": "JavaScript Basics",
    "tasks": [
      {
        "taskId": "t1",
        "title": "Learn variables and functions",
        "status": "completed"
      },
      {
        "taskId": "t2",
        "title": "Practice loops",
        "status": "pending"
      }
    ],
    "dueDate": "2026-07-05",
    "status": "pending"
  }
]

```

15.5 reminder_logs Table

```

CREATE TABLE reminder_logs (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID NOT NULL,
  roadmap_id UUID NOT NULL,
  week_number INT,
  type TEXT,
  email_sent BOOLEAN DEFAULT FALSE,
  sent_at TIMESTAMP DEFAULT NOW()
);

```

15.6 rag_knowledge_base Table with pgvector

```
CREATE TABLE rag_knowledge_base (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  title TEXT NOT NULL,  
  category TEXT,  
  content TEXT NOT NULL,  
  embedding VECTOR,  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

This table will store roadmap content, job role requirements, interview topics, ATS keywords, and learning resources for RAG-based retrieval.

16. APIs

Auth APIs

```
POST /api/auth/register  
POST /api/auth/login  
POST /api/auth/send-otp  
POST /api/auth/verify-otp  
POST /api/auth/forgot-password  
POST /api/auth/reset-password
```

Onboarding APIs

```
POST /api/onboarding/resume-upload  
POST /api/onboarding/auto-fill  
PUT /api/onboarding/profile  
GET /api/onboarding/profile
```

Skill Gap APIs

```
POST /api/skill-gap/analyze  
GET /api/skill-gap/:userId
```

Roadmap APIs

```
POST /api/roadmap/generate  
GET /api/roadmap/:userId  
POST /api/roadmap/task/complete  
GET /api/roadmap/progress/:userId
```

Reminder APIs

```
POST /api/reminders/check-weekly  
GET /api/reminders/logs/:userId
```

ATS Resume APIs

```
POST /api/resume-builder/generate  
GET /api/resume-builder/preview/:id  
GET /api/resume-builder/download/pdf/:id  
GET /api/resume-builder/download/docx/:id
```

17. Final Product Positioning

CareerGuid AI is not just a resume analyzer. It is a complete AI career preparation platform.

It helps users:

Upload resume



Auto-fill profile



Edit or complete missing details



Understand current skills



Find missing skills



Get personalized roadmap



Track roadmap progress



Receive weekly reminders



Build ATS-friendly resume



Become job-ready

18. One-Line Resume Description

CareerGuid AI - AI-Powered Career Guidance Platform Designed a scalable AI-powered career platform with resume-first smart onboarding, AI-based profile auto-fill, skill gap analysis, RAG-based roadmap generation, weekly progress reminder emails, GenAI career agent, ATS-friendly resume builder, Supabase PostgreSQL/Storage, Redis-based OTP/rate limiting, queue-based notifications, Docker deployment, and system design architecture.